

Traceability Matrix

Restaurant Management System

Team 10 - Software Testing Project 2026

Document Version	1.0
Date	May 29, 2026
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SRS Version	Final
Project	Restaurant Management System

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Last Updated: May 29, 2026

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1 Introduction

This Traceability Matrix maps each requirement from the SRS to the corresponding test cases. It ensures:

- All requirements have at least one test case
- Test coverage can be measured
- Missing or untested requirements are identified
- Changes to requirements can be traced to affected tests

1.1 Status Legend

Status	Meaning
Pending	Test case written but not yet executed
Pass	Test case executed and passed
Fail	Test case executed and failed (incident report created)
Blocked	Test case could not be executed (environment issue, etc.)
N/A	Requirement not testable (deferred to assumptions)

2 User Authentication Requirements (SRS Section 7.1)

Requirement ID	Requirement Description	Test Case ID(s)	Status
REQ-7.1.1	Customer registration with email, password, name, phone number	TC-RES-001, TC-RES-002	Pending
REQ-7.1.2	Authenticate users and create sessions upon successful login	TC-RES-003, TC-RES-004	Pending
REQ-7.1.3	Lock accounts for 15 minutes after three failed login attempts	TC-RES-005	Pending

3 Table Reservation Requirements (SRS Section 7.2)

Requirement ID	Requirement Description	Test Case ID(s)	Status
REQ-7.2.1	Display available time slots based on date, time, party size	TC-RES-006, TC-RES-007, TC-RES-008, TC-RES-009	Pending
REQ-7.2.2	Automatically assign tables based on party size and capacity	TC-RES-011, TC-RES-012, TC-RES-013	Pending
REQ-7.2.3	Allow pre-order selection during reservation with special instructions	TC-RES-015	Pending
REQ-7.2.4	Create reservations with Pending Approval status requiring staff confirmation	TC-RES-014, TC-RES-016, TC-RES-017	Pending
REQ-7.2.5	Allow customers to cancel reservations at least 24 hours in advance	TC-RES-018, TC-RES-019	Pending

4 Online Ordering Requirements (SRS Section 7.3)

Requirement ID	Requirement Description	Test Case ID(s)	Status
REQ-7.3.1	Display menu items organized by categories with details and pricing	TC-ORD-001, TC-ORD-002	Pending
REQ-7.3.2	Maintain shopping cart state during user sessions	TC-ORD-003, TC-ORD-004, TC-ORD-005, TC-ORD-006, TC-ORD-007	Pending
REQ-7.3.3	Validate delivery addresses against configured service zones	TC-ORD-010, TC-ORD-011	Pending
REQ-7.3.4	Process orders and generate unique order numbers with confirmation	TC-ORD-014, TC-ORD-015	Pending
REQ-7.3.5	Track order status through Received, Preparing, Ready, and Completed stages	TC-ORD-016, TC-ORD-017, TC-ORD-018, TC-ORD-019, TC-ORD-020	Pending

5 Event Booking Requirements (SRS Section 7.4)

Requirement ID	Requirement Description	Test Case ID(s)	Status
REQ-7.4.1	Check event space availability based on date, time, and guest capacity	TC-EVT-002, TC-EVT-007	Pending
REQ-7.4.2	Calculate catering costs based on guest count and selected menu items	TC-EVT-005	Pending
REQ-7.4.3	Prevent double-booking of event spaces through conflict detection	TC-EVT-003	Pending
REQ-7.4.4	Allow staff to approve or reject event requests with customer notifications	TC-EVT-009, TC-EVT-010, TC-EVT-011	Pending

6 Administrative Functions (SRS Section 7.5)

Requirement ID	Requirement Description	Test Case ID(s)	Status
REQ-7.5.1	Allow administrators to create, edit, and delete menu items and categories	TC-ADM-004, TC-ADM-005, TC-ADM-006, TC-ADM-007	Pending
REQ-7.5.2	Allow table and event space configuration with capacity settings	TC-ADM-008, TC-ADM-009	Pending
REQ-7.5.3	Allow administrators to create staff accounts with role assignment	TC-ADM-010	Pending

7 Non-Functional Requirements (SRS Section 8)

Requirement ID	Requirement Description	Test Case ID(s)	Status
REQ-8.1.1	Response Time: All user operations complete within 3 seconds	Manual timing during execution	Pending
REQ-8.2.1	Data Integrity: Zero double-booking incidents	TC-RES-010, TC-EVT-003	Pending
REQ-8.3.1	Authentication: All protected features require valid user authentication	TC-LGN-003, TC-LGN-004, TC-LGN-005	Pending
REQ-8.3.2	Authorization: Role-based access control prevents unauthorized access	TC-LGN-006, TC-LGN-007	Pending
REQ-8.3.3	Session Security: Sessions timeout after 15 minutes of inactivity	TC-RES-022	Pending
REQ-8.3.4	Data Protection: Passwords stored using secure hashing	Assumed (ASM-07)	N/A

8 Cross-Module & Workflow Requirements

Requirement ID	Requirement Description	Test Case ID(s)	Status
REQ-10.1	Customer makes reservation with pre-order workflow	TC-RES-006 to TC-RES-017	Pending
REQ-10.2	Customer places online order workflow	TC-ORD-001 to TC-ORD-025	Pending
REQ-10.3	Customer books event workflow	TC-EVT-001 to TC-EVT-012	Pending

9 Coverage Summary

Requirements Coverage

SRS Section	Requirements	Test Cases Assigned
Section 7.1 - User Authentication	3	5
Section 7.2 - Table Reservation	5	15
Section 7.3 - Online Ordering	5	25
Section 7.4 - Event Booking	4	7
Section 7.5 - Administrative Functions	3	6
Section 8 - Non-Functional	6	6 (plus assumptions)
Section 10 - Workflows	3	All test cases
Total	29	78

Coverage Percentage

Category	Coverage	Status
Functional Requirements	100% (29 of 29)	All have test cases
Non-Functional Requirements	83% (5 of 6)	Password hashing assumed
Workflow Requirements	100% (3 of 3)	Covered by functional tests
Overall Coverage	95%	Target achieved (more than 80%)

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